Phloton α

enhanced innovations

Reliable cold chain. Reliable healthcare delivery.

For pharma, labs, blood banks, clinics — and the patient

Proven in India · Built for the world

Majlis Venture Demo Day, powered by IRENA

Abu Dhabi · 17 October 2026



2 L units in assembly, Bengaluru

This region led the world's vaccine logistics when it needed it the most. Now let's take it to the last mile.

250M+

doses delivered to 60+ countries by Abu Dhabi's Hope Consortium

AD Ports / Maqta Gateway, 2022

$34B

UAE healthcare spend by 2027

DHCA, Arab Health 2024

38%

of pharma products now need cold chain — 26% in 2017

IQVIA, 2025

Half of what the world makes never protects anyone.

50%

of vaccines wasted every year at WHO's upper estimate; up to a quarter lost to temperature alone

WHO, via UNEP 2020 · IQVIA 2025

$35B

lost by biopharma yearly to temperature failures

IQVIA Institute, 2019

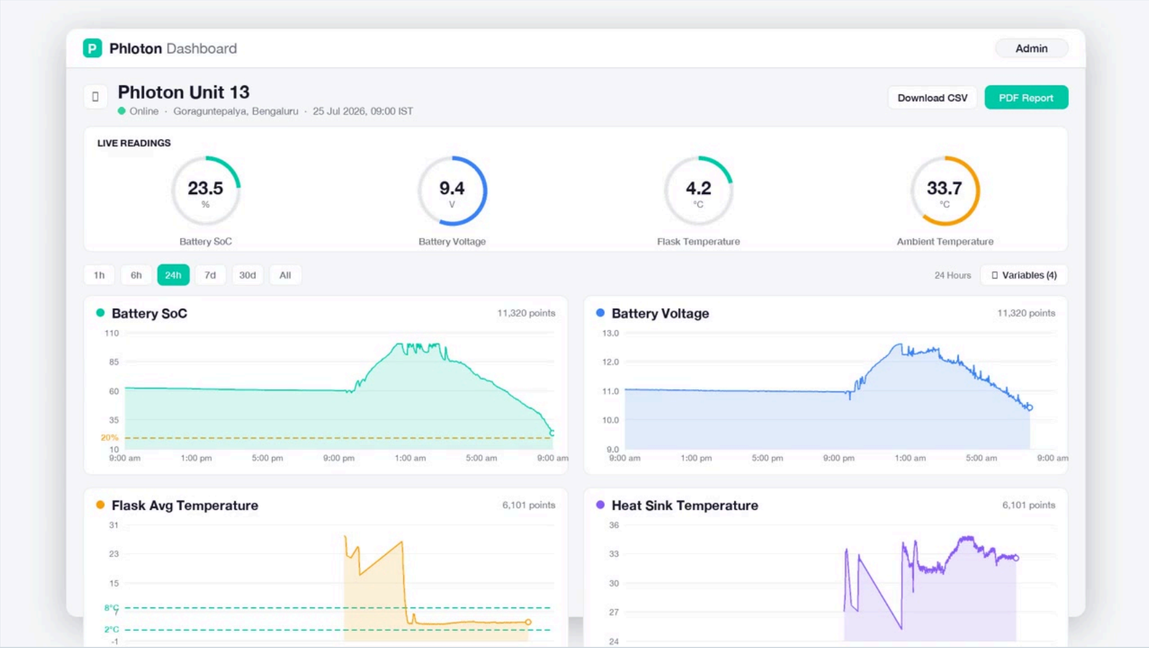
1B

people in lower-income countries served by health facilities without reliable power

WHO · World Bank · IRENA, 2023

2–8 °C. Sixteen hours at 43 °C. Charged by the sun.

AMBIENT	PAYLOAD	HOLD	AUDIT TRAIL
43.0 °C	3.0 °C	16 h	Live



The dashboard — 16 sensing points, alerts, fleet and location map



The 2 L unit

One engine. Every payload that must stay cold.

PAYLOAD	HOLD BAND	WHERE IT STANDS
PROVEN IN THE FIELD		
Vaccines	2–8 °C	SELCO Foundation outreach clinics · PCM jackets for open-vial sessions
Blood bags	2–6 °C · never freezes	Raktamitra, GIMS Noida — transported with no haemolysis
Diagnostic samples	2–8 °C	Beckman Coulter — six months of 7–8 h runs, then a purchase
Specialty biologics	2–8 °C	Roche India, Kolkata — cold-chain-as-a-service, depot to hospital · starting
Veterinary & dairy	2–8 °C	Akshayakalpa · URMUL
NEXT — SAME ENGINE, NEW SET POINT OR FORM FACTOR		
Insulin & antivenom	2–8 °C	2.5 L for primary health centres
Reagents, enzymes, trial kits	programmable	5 L for labs and clinical-trial logistics
Organs & tissue	4–10 °C	10 L · exploratory — a segment with no active portable player

Active, off-grid, connected.
Only one does all three.

	ICE-PACK BOX	PCM BOX	BLACKFROG EMVOLIO	PHLOTON
PRICE	$600 + freezer	$740 + freezer	~$2,000	$700 → $500 at scale
HOLDOVER	melts in hours	fixed window, halves each opening	12 h	16 h at 43 °C · 24 h at 27 °C
WEIGHT · BATTERY	—	—	5.5 kg · 200 Wh	3.75 kg · 390 Wh
CHARGING	needs a freezer	needs a freezer	mains	solar, vehicle or mains
MONITORING	none	none	live tracking	16 sensing points · 4G with offline buffer · per-trip compliance report
WHO PQS	listed (E004)	varies	listed (E003)	on the E003 route — funded by this round

$700 device. $60 a year. 5-year warranty.

DEVICE	$700 per 2 L unit today, sold near cost at pilot volumes. At scale: $500 against a ~$200 build — 60% gross margin after enclosures are injection-moulded.
SERVICE	$60 per unit per year — telemetry, compliance reports, maintenance. ~14% of revenue by FY30.
CONTRACTS	Cold-chain-as-a-service for institutions: we own the audit trail, they own the patient.

Fifty units in service.
All fifty bought.

40 bought commercially · **10** bought through grant programmes

96%

of **1,883** monitored hold-hours in 2–8 °C

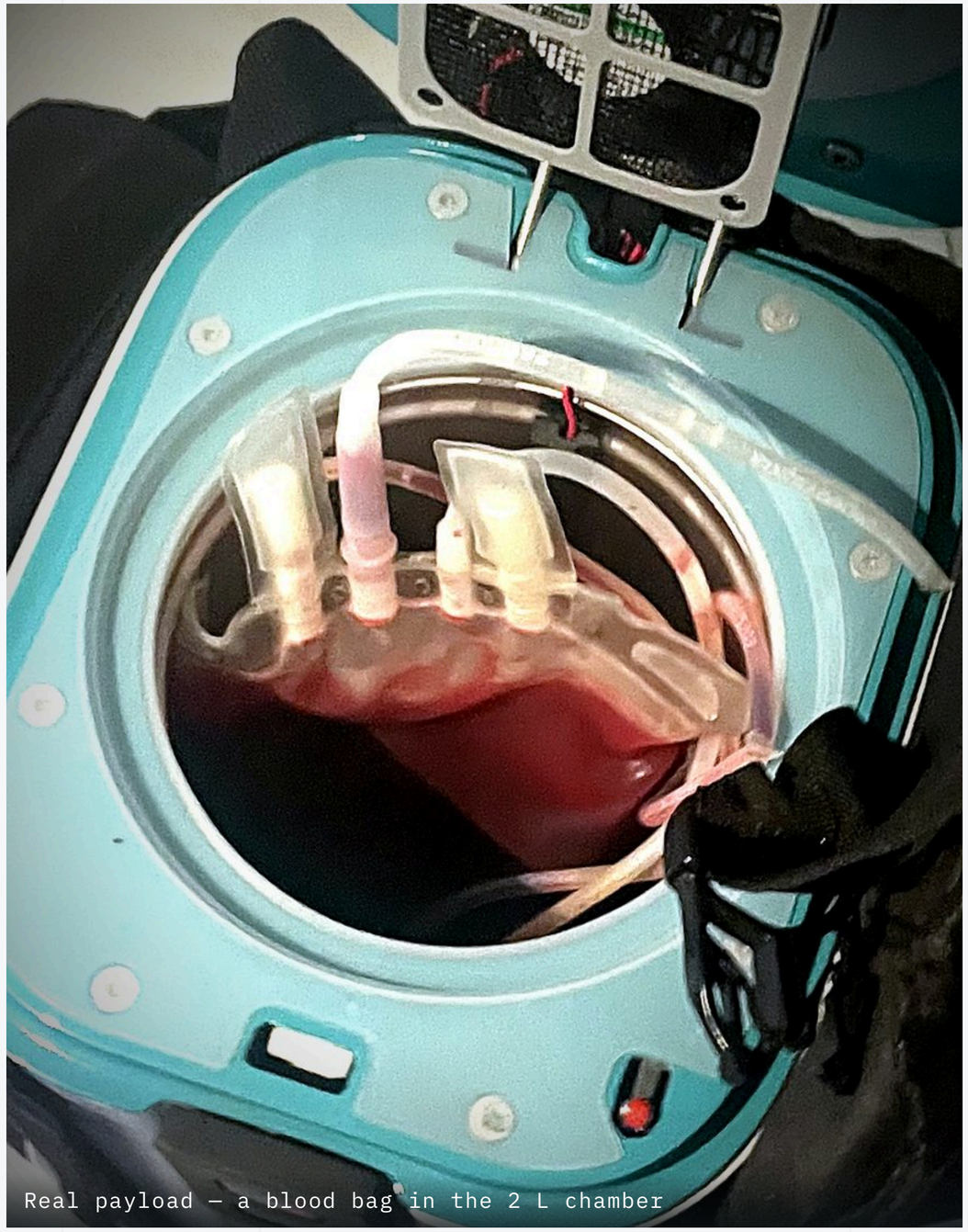
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readings below 0 °C, fleet-wide — nothing freezes

0

excursions in six months of 7–8 h sample runs at Beckman Coulter — then they bought

CLEARED	CDSCO test licence · TÜV SÜD EMC (IEC 61326-1) and electrical safety (IEC 61010-1) · IP55 (NABL) · patent and PCT filed · ISO 13485 auditor in-house
NEXT 12 MO	ISO 13485 + CDSCO MD-5 manufacturing licence · WHO PQS E003 submission



Real payload — a blood bag in the 2 L chamber

Who buys it. Who backs it.

CUSTOMERS, PILOTS & DEPLOYMENTS

Beckman Coulter — Diagnostics	Roche — Cold-chain-as-a-service · Kolkata	SELCO Foundation — PHC outreach	SVYM Swami Vivekananda Youth Movement — District hospital	Akshayakalpa — Dairy & veterinary
Disha Eye Hospitals (Open Vision Together) — Eye care	Urmul Trust — Rural health, Rajasthan	TIGS Tata Institute for Genetics and Society — Genetics research	Radical — Radical Health · screening	Solumatix — 30-unit order

INCUBATED & FUNDED BY

बाइरैक इग्नाइट इनोवेट इंक्यूबेट birac Ignite Innovate Incubate — SEED fund	MeitY Startup Hub Delivering Innovation! — TIDE 2.0	DST NIDHI START-TO-SCALE STARTUP SUPPORT — NIDHI PRAYAS	C-CAMP Centre for Cellular and Molecular Platforms — Incubation	IKP EDEN — Incubation
BFI BIOME NETWORK — Grant	Jubilant Bhartia Foundation — IQ 2.0 grant	Royal Academy of Engineering — LIF fellowship	#startupindia — DPIIT recognised	5 states — Karnataka · Tamil Nadu · Maharashtra · Gujarat · Uttar Pradesh

We need one percent of this market.

$1.2B

TAM — ~1M last-mile cold-chain devices × lifetime value

$0.7B

SAM — India, Middle East, Central Asia, Africa, SE Asia

$12M

SOM — 1% of TAM a year by year five; our plan reaches two-thirds of it. B2B for three years.

Made anywhere. Serviced by local technicians.

MAKE	Built in India today; assembled in-region wherever volume or policy asks for it. Solid-state design — no refrigerant handling, no compressor line.
SERVICE	Local technicians trained with every deployment; 4G diagnostics tell them what to fix before they travel.

3.2B

vaccine doses UNICEF delivered to 103 countries in 2025 — all of them cross a last mile

UNICEF Annual Results, 2025

13.5M

children received no vaccine at all in 2025

WHO / UNICEF, July 2026

1 in 2

hospitals in sub-Saharan Africa lack reliable electricity; 15% of facilities have none

WHO · World Bank · IRENA, 2023

1,900

doses protected per unit, per year

43M

doses a year by FY30 across 22,800 units

10,200t

CO_2e avoided a year by FY30 — no compressor, no refrigerant. Cooling already drives 7% of global emissions.

From a 50 ml prototype to a deployed platform.



2021

Genesis

50 ml solar-powered unit. Proof of concept.



2023

Field trials

10 units deployed with SELCO Foundation.



2024

Ver 3.0

Redesigned from field data. CDSCO test licence.



2025

Commercial-ready

Analytics on board. TÜV SÜD cleared. Patent filed.



2026 →

Scale

2.5 L, 5 L and 10 L. International expansion.

Today the device.
Tomorrow the chain.



The Morning Star · R&D centre → 10 L → cold container → 5 L, charged by the sun → 2 L backpack → her child, at sunrise · 60 s, with sound

Design, electronics and firmware. All developed in-house.



Suraj J R
CO-FOUNDER
Built the cold engine and next-gen controller

Shruti
FIRMWARE
Embedded and edge-AI control

Ankita Mittal
CEO
ISB · market entry, first institutional deployments

Prerit Mittal
FOUNDER
IIM-A · Royal Academy of Engineering Fellow

Chairman · Harkesh Kumar Mittal — architect of India's national incubation programme (NIDHI); chaired DPIIT's Start-up India seed fund committee

$1.1M at $8.5M pre-money.

One percent of the market. One round to get there.

35%	Go-to-market — UAE & India
25%	Team & field operations
25%	5 L / 10 L manufacturing
15%	IP, WHO PQS, working capital

Beyond capital

Pilot partners in the Gulf and Central Asia

Introductions to health authorities, distributors, DFIs

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What this round builds — the 5 L platform, FY27